



Barker
College

2021

YEAR 12
ASSESSMENT BLOCK

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Student Number

Mathematics Advanced

Staff Involved:

• AYG • LZM • LAK • DXC • PJR
• RJW* • PDJ • DZP • ARP

FRIDAY 26th MARCH, NOON

140 copies

TOPICS COVERED:

Year 11 Content	Sequences & Series
Series & Finance	Graphs & Equations
Curve Sketching using the Derivative	Integration

General

Instructions:

- Working time - 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A separate reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and / or calculations
- Marks may not be awarded for careless or poorly arranged working

Total marks:

80

Section I - 8 marks (pages 2 - 4)

- Attempt Multiple Choice Questions 1 - 8
- Allow about 15 minutes for this section

Section II - 72 marks (pages 5 - 22)

- Attempt Questions 9 - 17
- Show all necessary working
- Allow about 1 hours and 45 minutes for this section

Section I

8 marks

Attempt Questions 1 – 8

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 8.

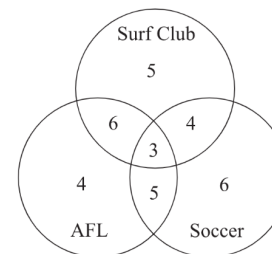
1 Using the log laws, fully simplify: $9 \log_a(\sqrt[3]{a})$

- A. 3
- B. 27
- C. $\sqrt[3]{9}$
- D. $3 \log_a(a)$

2 The sum of the first 10 terms of the series: $100 - 80 + 64 - \dots$ is closest to:

- A. -55
- B. 50
- C. 55
- D. 446

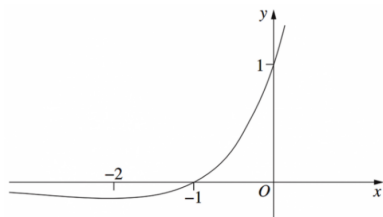
3 In a classroom, students are asked what sports club they are members of and the results are shown in the Venn diagram.



A student who is a member of a soccer club is chosen at random. What is the probability that he/she is also a member of a surf club?

- A. $\frac{2}{5}$
- B. $\frac{4}{11}$
- C. $\frac{2}{9}$
- D. $\frac{7}{18}$

- 4 The diagram shows the graph of $y = e^x(1+x)$



How many solutions are there to the equation $e^x(1+x) = 1-x^2$?

- A. 0
 B. 1
 C. 2
 D. 3
- 5 A circle with centre $(a, -2)$ and radius 5 units has equation $x^2 - 6x + y^2 + 4y = b$ where a and b are real constants.

The values of a and b are respectively:

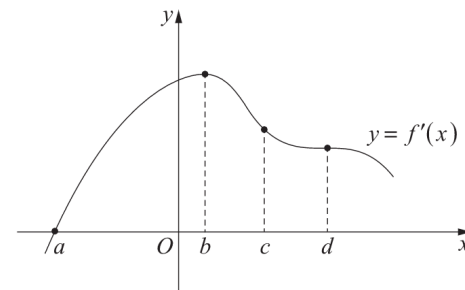
- A. -3 and 38
 B. 3 and 12
 C. -3 and -8
 D. 3 and 18

- 6 Let $f(x) = x^2$

Which one of the following is **not** true

- A. $f(ab) = f(a)f(b)$
 B. $f(a) - f(-a) = 0$
 C. $f(2a) = 4f(a)$
 D. $f(a-b) = f(a) - f(b)$

- 7 The diagram shows the graph of $f'(x)$, the derivative of a function.



For what value of x does the graph of the function $f(x)$ have a point of inflexion?

- A. $x = a$
 B. $x = b$
 C. $x = c$
 D. $x = d$
- 8 The function $y = f(x)$ has a turning point at $(4, 2)$. It is transformed to $y = -2f(2x+6) + 3$. Where is the new turning point?
- A. $(-1, -1)$
 B. $(-5, -2)$
 C. $(-5, -1)$
 D. $(1, -1)$

End of Section I

Student Number

Allow about 1 hour and 45 minutes for this section

In Questions 9 – 17, your responses should include relevant mathematical reasoning and/or calculations.

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Student Number

Question 10 (8 marks)

On any given day, the number X of telephone calls that Luke receives is a random variable with probability distribution given in the table below.

2

- (i) Show that $p = 0.1$ and hence complete the discrete probability distribution table, assuming no other values of x are possible.

x	0	1	2	3	4	5
$P(X = x)$	$p =$	0.15	$2p =$	$3p =$	0.2	0.05

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- (ii) Find $E(X)$.

2

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- (iii) What is the probability that Luke receives more than the mean number of phone calls on a given day?

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- (iv) Calculate the $Var(X)$.

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- (v) Calculate the standard deviation, to 2 decimal places.

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Student Number

Question 11 (8 marks)

- (a)
- (i) Sketch the curve $f(x) = (x - 1)(x + 1)(x + 3)$, showing x and y intercepts, in the space below. No need to find turning points. 2

- (ii) Hence or otherwise, solve $f(x) \leq 0$, giving your answer in interval notation. 2
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Question 11 (continued)

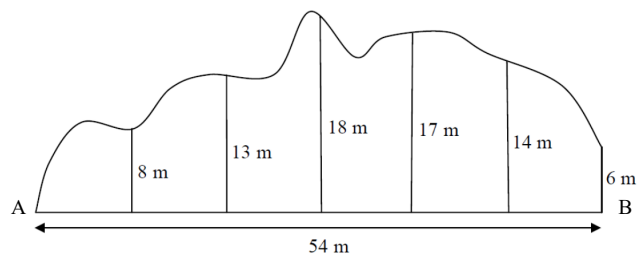
- (b) Evaluate the following given that $f(x) = \sqrt{1 - x}$ and $g(x) = \frac{1}{x}$
- (i) $g(f(-3))$ 1
-
-
-
-
- (ii) Determine the domain and range of $g(f(x))$ 3
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Question 12 (8 marks)

Student Number

- (a) The sketch shows a piece of land which borders the side of a straight road AB. 3

The length of AB is 54m. At equal intervals along AB, perpendicular measurements are made to the boundary as shown in the sketch.



Use the trapezoidal rule with 7 function values to find the approximate area of the piece of land.

[illegible]

Question 12 (continued)

- (b) (i) Prove the identity $(1 - \sin \theta)(1 + \sin \theta) = \cos^2 \theta$ 2

[illegible]

- (ii) Hence solve the equation $(1 - \sin \theta)(1 + \sin \theta) = 0.5$ for $0^\circ \leq \theta \leq 360^\circ$ **3**

[illegible]

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Student Number

Question 13 (8 marks)

Consider the curve $y = x^3 - \frac{3x^2}{2} - 6x + 5$

- (i) Find the stationary points and determine their nature.

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- (ii) Find the co-ordinates of the point of inflection and prove it is a point of inflection. 2

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Question 13 (continued)

- (iii) Sketch the curve $y = x^3 - \frac{3x^2}{2} - 6x + 5$ in the interval $[-3, 4]$ in the space below, showing all important features. (No need to find the x -intercepts)

2

- (iv) Find the maximum value of $x^3 - \frac{3x^2}{2} - 6x + 5$ in the domain $[-3, 4]$.

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Question 14 (8 marks)

Student Number

- (a) State the transformations that have occurred to the following graphs from the originals. The order may or may not be important.

(i) From $y = \ln x$ to $y = -\ln(x + 3)$ 1

(ii) From $y = e^x$ to $y = 2e^x - 5$ 1

(b) The function $f(x) = |x|$ is transformed and the equation of the new function is $y = kf(x + b) + c$.

The graph of the new function is shown below.



Question 14 (continued)

- (c) (i) Use differentiation by first principles to find $f'(x)$ given $f(x) = 2x^2 + 5x$. **2**

(ii) Hence, find the equation of the tangent to the curve $y = 2x^2 + 5x$ at the point on the curve where $x = 1$. 2

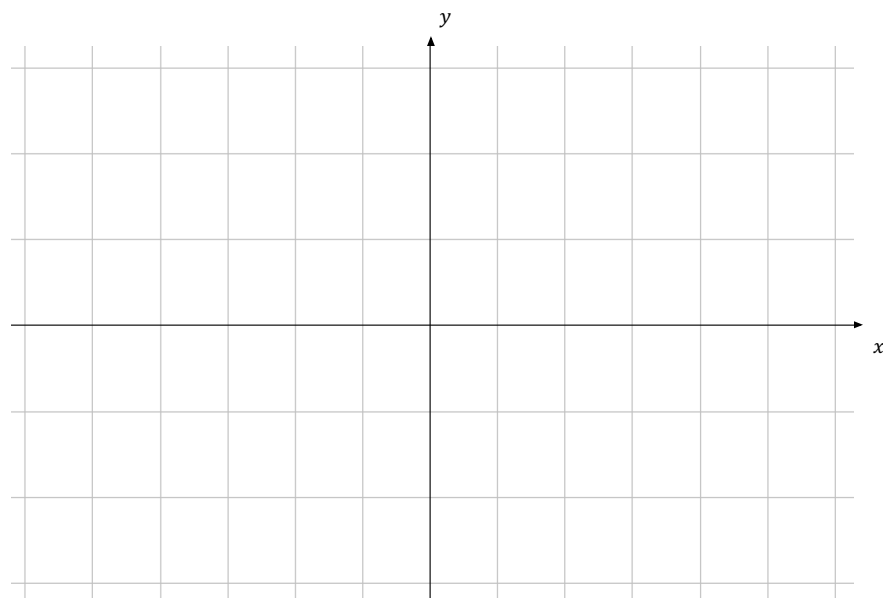
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Student Number

Question 15 (8 marks)

- (a) Sketch the curve $y = 2\cos x$ in the interval $[-2\pi, 2\pi]$. Label x and y intercepts. 2



Question 15 (continued)

- (b) Lucy borrows \$500 000 to be repaid at a reducible interest rate of 0.6% per month. Let A_n be the amount owing at the end of n months and M be the monthly repayment.

- (i) Show that $A_2 = 500000(1.006)^2 - M(1 + 1.006)$ 1

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- (ii) Show that $A_n = 500000(1.006)^n - M\left(\frac{1.006^n - 1}{0.006}\right)$ 2

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- (iii) Lucy budgets a monthly repayment of \$3500. Determine how many months it would take her to pay off the loan. 3

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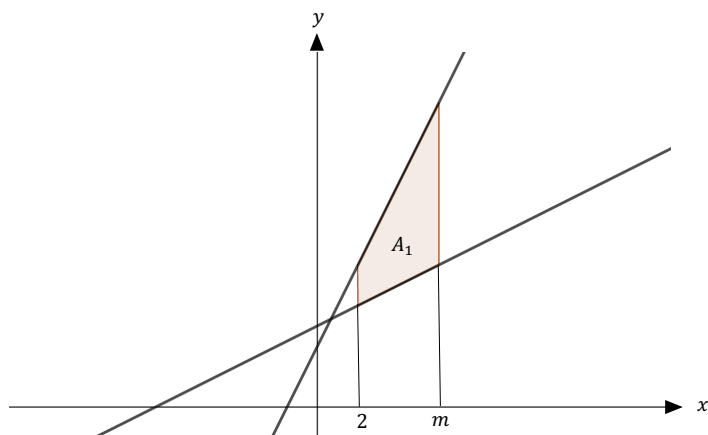
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Student Number

(a)

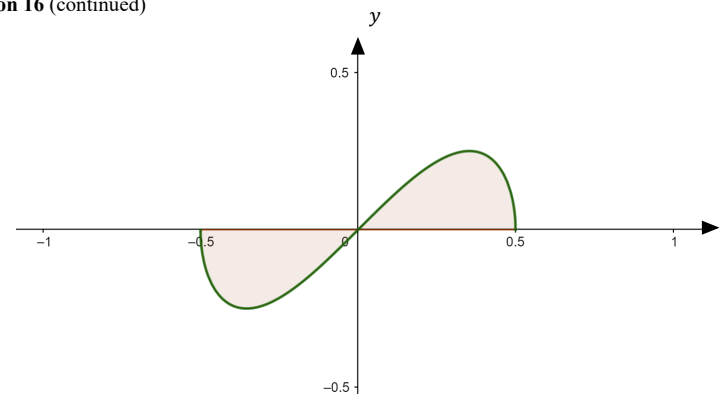


Given that $\text{area } A_1 = 20 \text{ units}^2$, find the value of m marked on the diagram.

3

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

(b)



(i) Find $\frac{d}{dx}(1 - 4x^2)^{\frac{3}{2}}$ and express your answer in surd form.

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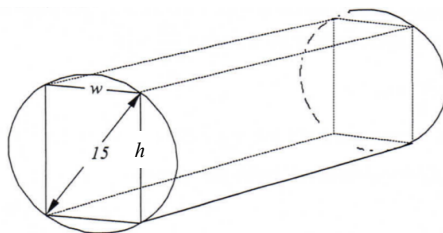
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3

[illegible]

Question 17 (8 marks)

Student Number



(i) Show that $S = k(225w - w^3)$

1

(ii) Hence, what dimensions will give a beam of maximum strength? Justify your answer. 4

4

Question 17 (continued)

- (iii) A square beam with a diagonal of 15 cm could have been cut from the log.

Show that the rectangular beam of maximum strength is more than 8% stronger than this square beam.

3

End of Paper

year 12 Advanced Ass #2 Solutions

26/03/21

Section 1

① $9 \log(a^{1/3})$
 $= \frac{1}{3} (9 \log a)$
 $= 3(1)$
 $= 3$

(A)

② $S_{10} = 100(1 - (-\frac{4}{5})^{10})$
 $1 + \frac{4}{5}$

$= 49.59 \dots$

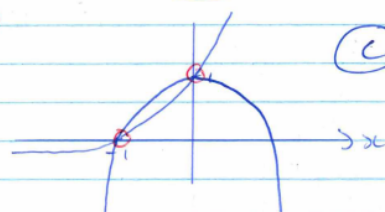
(B)

③ $7/18$

(D)

④

(C)



⑤ $x^2 + 6x + 9 + y^2 + 4y + 4 = 13 + b$
 $(x+3)^2 + (y+2)^2 = 13 + b$
 Centre $(-3, -2)$ $R = \sqrt{13+b}$
 $25 - 13 = b$
 $12 = b$

(B)

(D)

(B)

⑧ $y = -2f(2(x+3)) + 3$
 $(4, 2)$
 $x + \frac{1}{2} \rightarrow (4, -4) \rightarrow x - 2$
 $(2, -4)$
 $x - 3 \rightarrow (-1, -4) \rightarrow x - 3$
 $(-4, -1) \rightarrow x + 3$
 $(-1, 2)$

(A)

Section 2

④ (a) $\frac{dy}{dx} = 6(3x^2 - 4)^5 \times 6x$
 $= 36x(3x^2 - 4)^5$

(B) $u = x \quad v = e^x$
 $u' = 1 \quad v' = e^x$

$\frac{dy}{dx} = xe^x + e^x$
 $= e^x(x+1)$

(Y) $u = (x-1)^4 \quad v = x^2$
 $u' = 4(x-1)^3 \quad v' = 2x$

$\frac{dy}{dx} = \frac{4x^2(x-1)^3 - 2x(x-1)^4}{x^4}$
 $= \frac{2x(x-1)^3(2x - (x-1))}{x^4}$
 $= \frac{2(x-1)^3(x+1)}{x^3}$

9(b)(2) $3x^2 - \frac{x^4}{2} + C$

(B) $\frac{(3x-2)^{-3}}{-3x^3} + C$
 $= -\frac{(3x-2)^{-3}}{9} + C$

10(i) $p + 0.15 + 2p + 3p + 0.2 + 0.05 = 1$
 $6p = 0.6$
 $p = 0.1$

$\therefore 2p = 0.2 \quad 3p = 0.3$

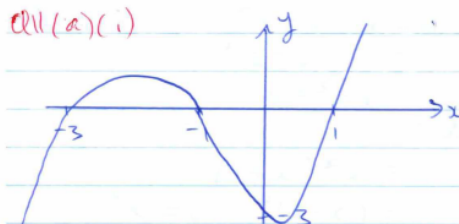
(ii) $E(X) = 2.5$

(iii) $P(X > 2.5) = 0.3 + 0.2 + 0.05$
 $= 0.55$

(iv) $\text{Var}(X) = 8 - 1 - (2.5)^2$
 $= 1.85$

(v) $\sigma = \sqrt{1.85}$
 $= 1.36$

11(a)(i)



(ii) $(-\infty, -3] \cup [-1, 1]$

(b)(i) $f(-3) = \sqrt{1+3} = 2$

$g(2) = \frac{1}{2}$

(ii) $g(\sqrt{1-x}) = \frac{1}{\sqrt{1-x}}$

D: $x < 1$

R: $y > 0$

12(a) $A \approx \frac{b-a}{2n} = \frac{54-0}{2(6)} = 4.5$

$A \approx 4.5(0 + 2(8+13+18+17+14) + 6)$
 $\approx 4.5(146)$
 $\approx 657 \text{ m}^2$

(b)(i) $1 - \sin^2 \theta = \cos^2 \theta$

From Ref Sheet $\sin^2 \theta + \cos^2 \theta = 1$
 $\cos^2 \theta = 1 - \sin^2 \theta$

$\therefore \cos^2 \theta = \cos^2 \theta$

(ii) $\cos^2 \theta = \frac{1}{2}$

$\cos \theta = \pm \frac{1}{\sqrt{2}}$

$\theta = 45^\circ, 135^\circ, 225^\circ, 315^\circ$

13(i) $y' = 3x^2 - 3x - 6$

$x^2 - x - 2 = 0$

$(x-2)(x+1) = 0$

$x = 2, -1$

$x = 2 \quad y = -5 \quad | \quad x = -1 \quad y = 8.5$
 $(2, -5) \quad | \quad (-1, 8.5)$

$y'' = 6x - 3$
 $x = 2$

$y'' = 6(2) - 3$
 $= 9 > 0$ C. up
 $(2, -5)$ L. min

$x = -1$

$y'' = 6(-1) - 3$
 $= -9 < 0$ C. down
 $(-1, 8.5)$ L. max

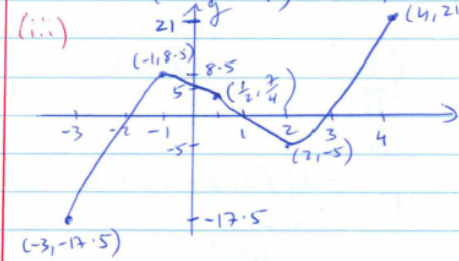
(ii) $6x - 3 = 0$
 $x = \frac{1}{2}$

$y = 7/4$

x	0	1/2	1
y''	-3	0	3

 $< 0 \quad > 0$

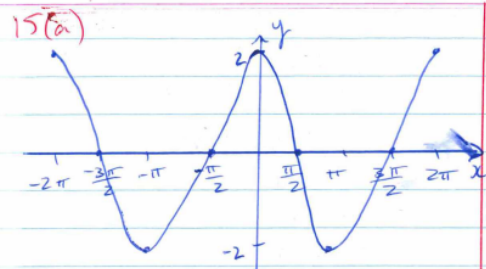
$\therefore$ Change in concavity
 $(1/2, 7/4)$ P.O.T



(iv) $y = 21$

14(a)(i) Reflect x-axis, left 3
(ii) Vertical dilation by 2,
down 5 spaces

(b) $k = -1/3$
 $b = 3$
 $c = 2$



(b) (i) $A_1 = 500000(1.006) - m$
 $A_2 = (500000(1.006) - m)(1.006) - m$
 $= 500000(1.006)^2 - m(1.006) - m$
 $= 500000(1.006)^2 - m(1 + 1.006)$

(ii) $A_N = 500000(1.006)^N - m(1 + 1.006 + \dots + 1.006^{N-1})$
 $= 500000(1.006)^N - m \left(\frac{1(1.006^N - 1)}{1.006 - 1} \right)$

$A_N = 500000(1.006)^N - m \left(\frac{1.006^N - 1}{0.006} \right)$

(iii) $0 = 500000(1.006)^N - 3500 \left(\frac{1.006^N - 1}{0.006} \right)$
 $3500 \left(\frac{1.006^N - 1}{0.006} \right) = 500000(1.006)^N$

$3500(1.006^N) - 3500 = 3000(1.006)^N$

$3500(1.006)^N - 3000(1.006)^N = 3500$

$500(1.006)^N = 3500$

$(1.006)^N = 7$

$N = \ln(7)$

$\ln(1.006)$

$N = 325.29 \dots$

$N = 326$

16(a) $\int_2^m (2x + 3 - \frac{1}{2}x - 4) dx = 20$

$= \int_2^m (\frac{3}{2}x - 1) dx = 20$

$= \left[\frac{3x^2}{4} - x \right]_2^m = 20$

$\frac{3m^2}{4} - m - \left(\frac{3(2)^2}{4} - 2 \right) = 20$

$\frac{3m^2}{4} - m - 1 = 20$

$3m^2 - 4m - 84 = 0$

$(3m + 14)(m - 6) = 0$

$m = \frac{-14}{3}$ $m = 6$

(b) (i) $\frac{dy}{dx} = \frac{3}{2}(1 - 4x^2)^{1/2}(-8x)$
 $= -12x\sqrt{1 - 4x^2}$

(ii) $2 \left(\frac{-1}{12} \right) \int_0^{0.5} -12x\sqrt{1 - 4x^2} dx$
 $= -\frac{1}{6} \int_0^{0.5} (1 - 4x^2)^{3/2} dx$

$= -\frac{1}{6} \left[(1 - 4(0.5)^2)^{3/2} - (1 - 4(0)^2)^{3/2} \right]$

$= -\frac{1}{6} (0 - 1)$

$= \frac{1}{6}$

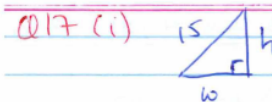
OR $\int_0^{0.5} x\sqrt{1 - 4x^2} dx$

$= 2 \int_0^{0.5} \frac{x(1 - 4x^2)^{3/2}}{(-8x)(3/2)} dx$

$= \frac{2}{-12} \left[(1 - 4(0.5)^2)^{3/2} - (1 - 4(0)^2)^{3/2} \right]$

$= \frac{2}{-12} (0 - 1)$

$= \frac{1}{6}$



$225 = w^2 + h^2$

$225 - w^2 = h^2$

$S = k(225 - w^2)(w)$

$= k(225w - w^3)$

(ii) $S = 225kw - kw^3$

$S' = 225k - 3kw^2$

$225k - 3kw^2 = 0$

$3kw^2 = 225k$

$w = \pm 5\sqrt{3}$

$S'' = -6kw$

$w = 5\sqrt{3}$

$= -6k(5\sqrt{3})$

$= -30k\sqrt{3}$

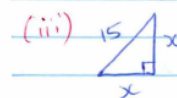
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$w = 5\sqrt{3}$ is L. Max

$225 - (5\sqrt{3})^2 = d^2$

$150 = d^2$

$d = 5\sqrt{6}$



$2x^2 = 225$

$x^2 = \frac{225}{2}$

$x = \frac{15\sqrt{2}}{2}$

Rectangle: $S = k(5\sqrt{6})^2(5\sqrt{3})$
 $= 750\sqrt{3}k$

Square:

$S = k \left(\frac{15\sqrt{2}}{2} \right)^2 \left(\frac{15\sqrt{2}}{2} \right)$

$= \frac{3375\sqrt{2}k}{4}$

$= \frac{750\sqrt{3}k}{4}$

$= \frac{3375\sqrt{2}k}{4}$

$= \frac{750\sqrt{3}k}{4} \times \frac{4}{3375\sqrt{2}k}$

$= 1.08866 \dots$

$\therefore 8.866\% \text{ more}$